

## **Data Structures and Algorithms**

**Max: 30pts**

**Year 2 Computer and Software Engineering**

### **Question 1 [10pts]**

- a. What is the general idea of Divide-and-Conquer technique?
- b. Explain in details the classification of Data Structure

### **Question 2 [10pts]**

- a. Given a list of unordered  $n$  elements, find max and min using the straightforward algorithm
- b. What are Content of a Recursive Method

### **Question 3 [10pts]**

- a. Let  $\alpha$  be some constant, independent of the input array length  $n$ , strictly between 0 and  $1/2$ . What is the probability that, with a randomly chosen pivot element, the Partition subroutine produces a split in which the size of both the resulting subproblems is at least  $\alpha$  times the size of the original array?
- b. Define a Queue and write an algorithm for the operations on Queue?

### **Question 4 [10pts]**

- a. Demonstrate the selection sort results for each pass for the following initial array of elements.  
21 6 3 57 13 9 14 18 2
- b. Write a program that searches a value in a stored array using recursive linear search.

### **Question 5 [10pts]**

- a. Define a stack and write an algorithm for the operation on stack?
- b. Write a function that, given a pointer to a singly-linked list and a number  $k$ , returns the  $k$ th-to last element of the linked list (or a null pointer if no such element exists). How efficient is your solution, from a big-O perspective? As a challenge, see if you can solve this in  $O(n)$  time with only  $O(1)$  auxiliary storage space.

### **Question 6 [10pts]**

- a. Write an algorithm that searches a value in a stored array using linear search
- b. What are the operations on Linear Lists? Differentiate between using Arrays and Linked Lists for implementation of Linear Lists

\*\*\*\* Good Luck \*\*\*\*